

MICRONUTRIENTS-VITAMINS AND MINERALS

Vitamins and minerals are micronutrients required by the body to carry out a range of normal functions. However, these micronutrients are not produced in our bodies and must be derived from the food we eat.

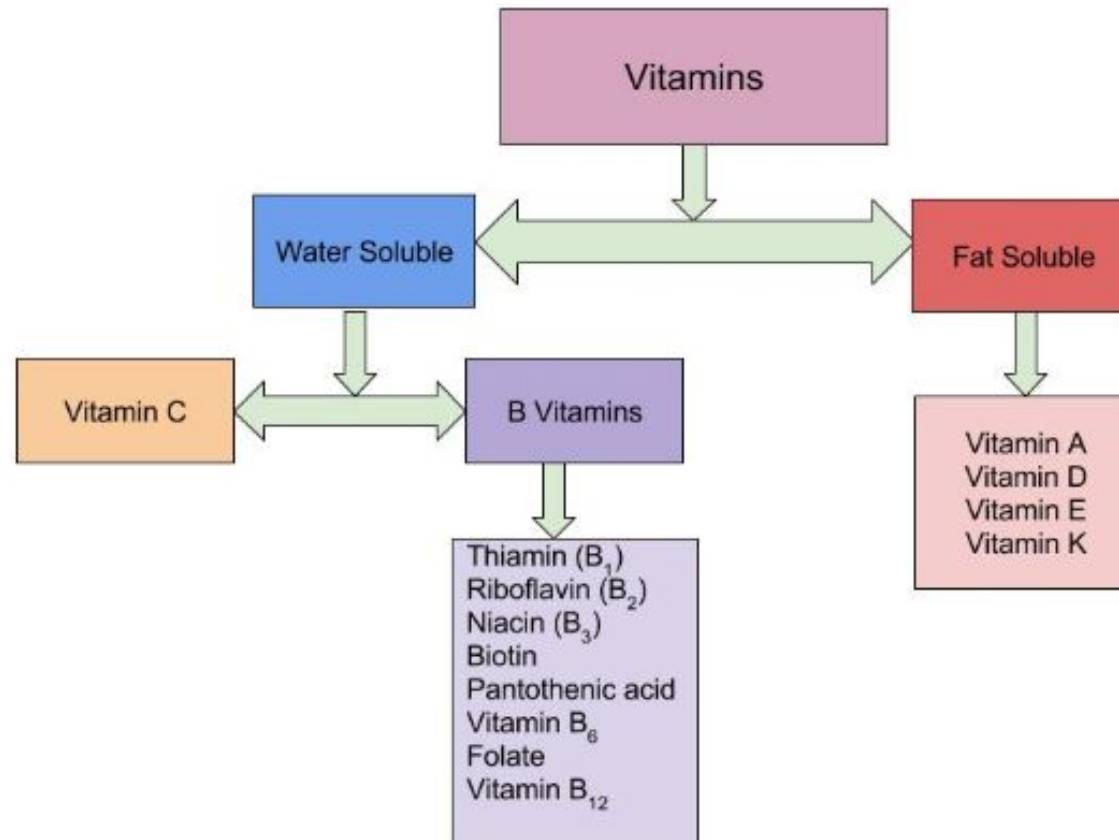
VITAMINS

A vitamin is an organic compound required in very small amounts in the diet that is essential for normal growth, metabolism, and maintenance of the body, and which the body cannot synthesize in sufficient quantities.

- They **do not provide energy** like carbohydrates, fats, or proteins. But they are **essential** for regulating body processes.

CLASSIFICATION OF VITAMINS

- Vitamins are traditionally categorized into two groups: **water-soluble** or **fat-soluble**.



CHARACTERISTICS OF FAT AND WATER SOLUBLE VITAMINS

Characteristics of Fat-Soluble Vitamins	Characteristics of Water-Soluble Vitamins
Protect cell membranes from free radical damage; act within the cell's nucleus to influence gene expression	Act in the cytosol of cells or in extracellular fluids such as blood
Absorbed into lymph with fats from foods	Absorbed directly into blood
Large storage capacity in fatty tissues	Little to no storage capacity
Do not need to be consumed daily to prevent deficiency (may take months to develop)	Need to be consumed regularly to prevent deficiency
Toxicity is more likely	Toxicity is rare

FAT SOLUBLE VITAMINS

- Fat-soluble vitamins are absorbed with the other fats from the food you eat. When they're absorbed in this way, they go with the fats and are stored in your body's fat tissue and liver.
- Fat-soluble vitamins can stay in your body for a while. When they're stored in your body's fat, they can be stored for up to 6 months until your body needs them.
- The primary fat-soluble vitamins in your diet are **vitamins A, D, E, and K**. Each one has its important function and can be found in a variety of food.

FAT SOLUBLE VITAMINS

VITAMIN A

- Vitamin A is a nutrient the body uses to support growth, vision and cell function.
- It's also called **retinol or retinoic acid**.
- Vitamin A has antioxidant properties. Antioxidants are substances that might protect cells against the effects of free radicals.
- Free radicals are molecules made from reactions in the body or when the body is exposed to tobacco smoke or radiation.
- Free radicals might play a role in heart disease, cancer and other conditions.

FAT SOLUBLE VITAMINS

Types of Vitamin A

- Vitamin A is a critical vitamin that helps your body develop and function properly. There are two types of vitamin A, which come from different sources:
- **Carotenoids (provitamin A)** are found in plant-based foods, such as carrots and sweet potatoes, fortified foods (where vitamins are added) and supplements. To digest carotenoids, your body needs healthy fats to change (convert) them into the other, active form of vitamin A (retinol). One of the most common carotenoids is beta-carotene, which is responsible for giving plants their orange and red pigments.
- **Retinoids (retinol or preformed vitamin A)** are found in animal-based foods, such as eggs, fish, milk and liver. Your body can use this form of vitamin A right away.

FAT SOLUBLE VITAMINS

Functions of Vitamin A

1. Maintains healthy vision

- One of vitamin A's most important roles is to preserve and maintain your vision. It helps change the light that hits your eye into an electrical signal that can be sent to your brain.
- Your body also uses vitamin A to make pigments for your retinas to work well, and moisture for your corneas.

2. Aids immune system function

- Vitamin A strengthens your immune system by supporting white blood cells and the mucus membranes in your lungs, intestines and urinary tract. This helps you ward off infection and toxins (also called free radicals) that cause inflammation and disease.
- Giving vitamin A supplements to children with measles has been shown to reduce the severity of the disease, according to the World Health Organization. In other words, vitamin A can sometimes save lives.

FAT SOLUBLE VITAMINS

3. **Reduces cancer risk**

- Vitamin A plays a key role in the healthy growth and development of your cells.

4. **Keeps skin clear**

- Vitamin A helps maintain healthy skin by supporting the growth and repair of skin cells. It prevents dryness and roughness by keeping the skin tissues moist and well-nourished. Vitamin A also helps in reducing acne and skin infections by strengthening the protective barrier of the skin.

5. **Supports reproductive health**

- Adequate amounts of vitamin A in your diet are essential for healthy reproductive function. Vitamin A deficiency can lead to infertility and cause delayed growth and development in children.
- But too much vitamin A during pregnancy can harm the fetus, causing birth defects and increased risk of infection and disease.

6. **Keeps bones and teeth strong**

- Vitamin A helps maintain proper bone growth and development, lowering your risk of injury or disability. But it's important to strike a balance.

FAT SOLUBLE VITAMINS-VITAMIN A

Recommended Dietary Allowance (RDA)

In India, the Recommended Dietary Allowance (RDA) for **Vitamin A** (as per the *Indian Council of Medical Research – National Institute of Nutrition* recommendations) depends on age and gender.

- RDA of Vitamin A for Indians (per day)
- Adult Men: 1000 µg retinol activity equivalents (RAE) per day
- Adult Women: 840 µg RAE per day

SOURCES OF VITAMIN A

Sources of Vitamin A come in two main forms:

- **Preformed Vitamin A (Retinol)** – from animal foods (readily absorbed)
- **Provitamin A Carotenoids (β-carotene)** – from plant foods (converted to vitamin A in the body)

FAT SOLUBLE VITAMINS-VITAMIN A

- **Plant Sources (Rich in β -Carotene)**

Vegetables

- Carrot, Pumpkin, Sweet potato, Spinach (palak), Drumstick leaves, Coriander leaves, Fenugreek leaves.

Fruits

- Mango, Papaya, Jackfruit, Apricot, Orange.

DEFICIENCY OF VITAMIN A

Vitamin A deficiency mainly affects the **eyes, skin, and immunity**. It is a major cause of **preventable blindness**, especially in children.

1. Eye Manifestations (Xerophthalmia sequence)

Night blindness (Nyctalopia) – earliest symptom; difficulty seeing in dim light

Conjunctival xerosis – dryness of conjunctiva

Bitot's spots – foamy white patches on conjunctiva

Corneal xerosis – dry cornea

Keratomalacia – softening and ulceration of cornea → may lead to blindness

2. Skin Changes

- Dry, rough skin
- Follicular hyperkeratosis (toad skin)

DEFICIENCY OF VITAMIN A

3. Reduced Immunity

- Frequent infections (respiratory, gastrointestinal)
- Poor resistance to diseases

4. Growth and Development Issues

- Growth retardation in children
- Delayed bone development

5. Poor Wound Healing

- Wounds that do not heal well after injury or surgery may be linked to low vitamin A levels.
- This is because vitamin A promotes the creation of collagen, an important component of healthy skin.

EXCESS OF VITAMIN A

Too much vitamin A can be harmful. Taking a single large dose of 200,000 mcg or many large doses can cause:

- Nausea.
- Vomiting.
- Fatigue.
- Dizziness.
- Blurry vision.

Taking more than 10,000 mcg a day of oral vitamin A supplements long-term can cause:

- Balance issues.
- Liver damage.
- Headache.
- Nausea.
- Hair loss.
- Dry skin.
- Pain in the joints and bones.
- Rarely, death.

VITAMIN D

Vitamin D (Calciferol)

- Vitamin D is a fat-soluble vitamin essential for calcium and phosphorus metabolism, bone health, and many regulatory functions in the body.
- Unlike most vitamins, Vitamin D can be **synthesized in the skin** on exposure to sunlight.
- How much vitamin D the body makes depends on many things. These include the time of day you're in the sun, what season it is, where you live, and the color of your skin. Depending on where you live and your lifestyle, you might get little to no vitamin D from the sun during winter.

VITAMIN D

FORMS OF VITAMIN D

Form	Name	Source
Vitamin D₂	Ergocalciferol	Plants, fungi
Vitamin D₃	Cholecalciferol	Skin synthesis, animal foods

FUNCTIONS OF VITAMIN D

1. Calcium and Phosphorus Absorption

- Increases intestinal absorption of Ca^{2+} and PO_4^{3-}
- Maintains normal blood calcium level

2. Bone and Teeth Formation

- Promotes mineralization
- Prevents soft bones

3. Regulation of Parathyroid Hormone

- Works with PTH to maintain calcium balance

FUNCTIONS OF VITAMIN D

4. Muscle Function

- Prevents muscle weakness and falls

5. Immune System Support

- Enhances immune defense
- Reduces infection risk

6. Cell Growth and Differentiation

- Important in preventing abnormal cell growth

RECOMMENDED DAILY ALLOWANCE OF VITAMIN D

Age Group	RDA	1 IU of Vitamin D = 0.025 micrograms (µg) of cholecalciferol (D ₃)
Infants	400 IU/day	
Children & Adults	600 IU/day	
Elderly (>70 yrs)	800 IU/day	
Pregnant/Lactating	600 IU/day	

EXCESS OF VITAMIN D

Safety and side effects

Taken in typical doses, vitamin D is thought to be mainly safe.

But taking too much vitamin D in the form of supplements can be harmful and even deadly. Taking more than 4,000 IU a day of vitamin D might cause:

- Upset stomach and vomiting.
- Weight loss and not wanting to eat.
- Muscle weakness.
- Not being able to think clearly or quickly.
- Heart rhythm issues.
- Kidney stones and kidney damage.

DEFICIENCY OF VITAMIN D

- **In Children – Rickets**

- Bow legs
- Delayed teeth eruption
- Soft skull bones
- Poor growth

- **In Adults – Osteomalacia**

- Bone pain
- Muscle weakness
- Fragile bones

- **In Elderly – Osteoporosis**

- Reduced bone density
- Fractures

DEFICIENCY OF VITAMIN D

Risk Factors for Deficiency

- Lack of sunlight exposure
- Dark skin
- Covering clothes
- Indoor lifestyle
- Malabsorption disorders
- Kidney/liver disease

VITAMIN E

Vitamin E is a fat-soluble antioxidant essential for human health that protects cell membranes from damage by reactive oxygen species and plays a role in immune function, skin health, and vision. It encompasses a group of eight compounds, including four tocopherols and four tocotrienols.

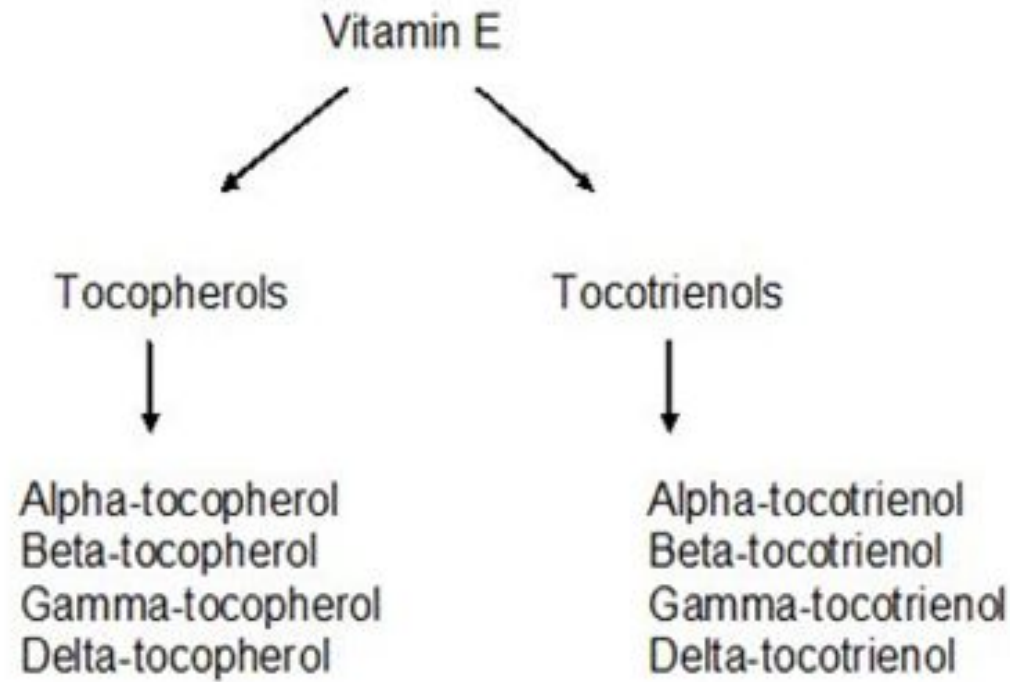
The most active form in humans is **α -tocopherol**.

- Vitamin E has the ability to protect cells from free radical damage as well as reduce the production of free radicals in certain situations.
- Antioxidant vitamins, including vitamin E, came to public attention in the 1980s when scientists began to understand that free radical damage was involved in the early stages of artery-clogging atherosclerosis, and might also contribute to cancer, vision loss, and a host of other chronic conditions.

VITAMIN E

Forms of Vitamin E

Vitamin E occurs in two major forms, **Tocopherols**, and **Tocotrienols**, with each form having 4 major groups (alpha, beta, delta, gamma).



VITAMIN E

Tocopherols

- Alpha Tocopherol : Major circulating vitamin E, antioxidant, protects LDL/membranes
- Beta Tocopherol : General antioxidant, less bioactive
- Delta Tocopherol : Anti-cancer, anti-inflammatory
- Gamma Tocopherol : Potent anti-inflammatory, scavenges peroxynitrite (RNS)

Tocotrienols

- Alpha Tocotrienols: Neuroprotection, antioxidant, micro-glial protection
- Beta Tocotrienols: General antioxidant, weak activity
- Delta Tocotrienols: Anti-cancer, anti-proliferative, mitochondrial (SDH/Cardiolipin)/metabolic regulation
- Gamma Tocotrienols: Anti-inflammatory, cholesterol-lowering, anti-cancer, mitochondrial (SDH/Cardiolipin) protection

VITAMIN E

Functions of Vitamin E

1. Antioxidant Property:

The primary function of vitamin E revolves around its antioxidant capability. It effectively prevents the non-enzymatic oxidations of various cellular components, such as unsaturated fatty acids, by molecular oxygen and free radicals like superoxide (O_2^-) and hydrogen peroxide (H_2O_2). Furthermore, the element selenium aids in enhancing these antioxidant functions.

2. Cell Membrane Integrity:

Vitamin E is pivotal for maintaining the structure and integrity of cell membranes. Therefore, it is often termed a membrane antioxidant.

3. Prevention of Red Blood Cell Hemolysis:

It shields red blood cells from hemolysis.

4. Reproductive Functions:

Vitamin E is intricately linked with reproductive processes, ensuring the prevention of sterility.

5. Delaying Cataract Onset:

In conjunction with vitamins A, C, and β -carotene, vitamin E helps in delaying the onset of cataracts.

VITAMIN E

6. **Cellular Respiration:** Vitamin E is essential for cellular respiration, particularly through the electron transport chain.

7. **Protection of Other Vitamins:**

Vitamin E safeguards vitamin A and carotenes from oxidation.

8. **Creatine Storage:**

It ensures the proper storage of creatine in skeletal muscles.

9. **Amino Acid Absorption:**

Vitamin E facilitates the optimal absorption of amino acids from the intestine.

10. **Nucleic Acid Synthesis:**

It plays a role in the proper synthesis of nucleic acids.

11. **Liver Protection:**

Vitamin E shields the liver from damage caused by toxic compounds, such as carbon tetrachloride.

12. **Prevention of Chronic Diseases:**

While vitamin E has been suggested for preventing chronic ailments like cancer and heart diseases, clinical trials have yielded mixed results. However, it is believed that vitamin E prevents the oxidation of LDL, which is implicated in promoting heart diseases.

VITAMIN E

Recommended Dietary Allowance (RDA):

Recommended Dietary Allowances (RDAs) for Vitamin E in India as per the latest Indian Council of Medical Research (ICMR) guidelines

Adults (men & women): 7.5–10 mg/day tocopherol.

DIETARY SOURCES OF VITAMIN E

- **Vegetable Oils:**

- Among the primary sources of vitamin E are vegetable oils. Wheat germ oil stands out as a particularly rich source. Besides wheat germ oil, other oils such as cottonseed oil, peanut oil, corn oil, and sunflower oil are also commendable sources of this nutrient.

- **Animal Products:**

- Animal-derived products also contribute to the dietary intake of vitamin E. Meat, for instance, contains a notable amount of this vitamin. Additionally, dairy products like milk and butter, as well as eggs, are also valuable sources.

VITAMIN E

- **Vegetables:**

- Vegetables are not to be overlooked when considering sources of vitamin E. Their inclusion in daily diets ensures a balanced intake of this essential vitamin.

- **Diverse Sources:**

- It's worth noting that soybean oil and cottonseed oil are also rich in vitamin E. These oils, along with previously mentioned sources like butter, milk, meat, and eggs, offer a diverse range of options for individuals to meet their vitamin E requirements.

VITAMIN E

DEFICIENCY OF VITAMIN E

Neuromuscular problems (most important)

- Peripheral neuropathy
- Muscle weakness (myopathy)
- Loss of reflexes
- Ataxia (unsteady gait)
- Difficulty in coordination and balance
- **Hemolytic anemia**
- RBC membrane becomes fragile
- RBCs break easily → anemia
- Common in **premature infants**
- **Retinopathy (vision problems)**
- Damage to retinal cells due to oxidative stress
- **Impaired immunity**
- Increased susceptibility to infections

VITAMIN E

EXCESS OF VITAMIN E

Vitamin E is fat-soluble, so excess intake (usually from **high-dose supplements**, not food) can accumulate and cause problems—mainly related to **bleeding**.

Bleeding tendency (most important)

- Vitamin E **antagonizes Vitamin K**
- Interferes with clotting factor synthesis
- Prolonged bleeding time
- Easy bruising, nosebleeds

Gastrointestinal disturbances

- Nausea
- Diarrhea
- Abdominal cramps

Fatigue and weakness

- General tiredness with very high doses

Headache and blurred vision

Increased risk of stroke (very high doses)

VITAMIN K

- Vitamin K is a fat-soluble vitamin essential for the synthesis of proteins involved in blood coagulation and bone health. It is naturally found in green leafy vegetables, certain animal foods, and is also produced by gut bacteria.

FORMS OF VITAMIN K

Form	Name	Source
K₁	Phylloquinone	Green leafy vegetables
K₂	Menaquinone	Intestinal bacteria, fermented foods
K₃	Menadione (synthetic)	Synthetic form

VITAMIN K

Phylloquinone (Vitamin K1):

- Phylloquinone, commonly referred to as Vitamin K1, is predominantly derived from plants. It plays a crucial role in photosynthesis, which is the process by which plants convert light energy into chemical energy.
- Therefore, green leafy vegetables, such as spinach, kale, and broccoli, are rich sources of Vitamin K1. Besides its role in plants, Vitamin K1 is active as a vitamin in animals, emphasizing its importance in the production of blood-clotting proteins.

Menaquinone (Vitamin K2):

- Menaquinone, or Vitamin K2, is not typically obtained directly from dietary sources. Instead, it is synthesized by bacteria present in the human gut. This form of Vitamin K is particularly intriguing because of its bacterial origin.
- Within the category of Vitamin K2, there are several subtypes, differentiated by the lengths of their carbon side chains. The most studied subtypes include menaquinone-4 (MK-4) and menaquinone-7 (MK-7). These subtypes further highlight the complexity and diversity of Vitamin K2.

Menadione (Vitamin K3):

- Menadione represents the synthetic form of Vitamin K. It was previously used to treat Vitamin K deficiencies. However, due to its interference with the function of glutathione, a critical antioxidant in the body, its use in human nutrition has been discontinued.

VITAMIN K

Recommended dietary allowance (RDA) of vitamin K

The Recommended Dietary Allowance (RDA) for vitamin K varies based on age, gender, and life stage. While the human gut can synthesize vitamin K with the help of bacteria, it's essential to obtain a portion of this vitamin from dietary sources to ensure optimal health.

For adults, the suggested RDA for vitamin K is as follows:

- Men (aged 19 years and older): 120 micrograms (µg) per day
- Women (aged 19 years and older): 90 micrograms (µg) per day

VITAMIN K

Dietary source of Vitamin K

- Vitamin K is a crucial nutrient that plays a pivotal role in various physiological functions, especially in the process of blood clotting. Therefore, understanding its dietary sources is essential for maintaining optimal health.
- **Green vegetables** are among the primary sources of vitamin K. Specifically, cabbage and cauliflower stand out as rich sources. Besides these, spinach, a leafy green vegetable, is also packed with this vital vitamin. Tomatoes further contribute to the dietary intake of vitamin K.
- However, vitamin K is not limited to plant-based sources. **Animal-derived foods**, such as egg yolk, are also enriched with this nutrient. Additionally, meat, particularly liver, contains significant amounts of vitamin K. Dairy products, including cheese and butter, are other notable sources.

VITAMIN K

Biochemical functions of vitamin K

Role in Blood Clotting: Vitamin K is instrumental in the blood clotting process

Bone formation and mineralization

- Activates **osteocalcin**, a protein needed to bind calcium to bone
- Helps in proper bone mineralization
- Prevents bone fragility

Role in calcium regulation

- Helps in directing calcium to bones and teeth
- Prevents abnormal calcium deposition in soft tissues and blood vessels

Cellular functions

- Involved in synthesis of certain proteins needed for cell growth and maintenance

VITAMIN K

Deficiency of Vitamin K

Vitamin K deficiency mainly affects the **blood clotting mechanism**, leading to a **bleeding tendency**.

Bleeding tendency (most important)

- Prolonged bleeding time
- Easy bruising
- Nosebleeds (epistaxis)
- Bleeding gums

Hemorrhagic disease of the newborn

- Common in newborn babies
- May lead to serious internal bleeding (including brain)

VITAMIN K

Toxicity is uncommon with natural forms (K_1 , K_2). Adverse effects are mainly linked to synthetic Vitamin K (K_3 – menadione) or high doses in infants.

Hemolytic anemia

- Destruction of RBCs → anemia

Jaundice (hyperbilirubinemia)

- From RBC breakdown
- Risk of kernicterus (serious brain damage) in newborns

Liver toxicity

- Especially with menadione (K_3)

Allergic reactions (rare)

- Skin rashes, irritation